

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T26 · Solving Inequalities

With Answers

Questions with full worked solutions

■ 50 pages

FAMILY MAP

Solving Inequalities

F01	Solve linear inequalities	10 Q 27 marks
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F02	Solve quadratic inequalities	5 Q 17 marks
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F03	Translate inequality notation and number lines	7 Q 27 marks
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9 (a) Solve the inequality $7 - 3t < 2t + 15$

(2)

Worked Solution 24

Linear inequality

UNIT 2
2 marks**Collect terms**

$$7 - 3t < 2t + 15 \Rightarrow -5t < 8.$$

Reverse on division

$$t > -\frac{8}{5} = -1.6 \quad \text{because division is by } -5.$$

Final answer

$$t > -1.6$$

- 19** Solve the inequality $4x^2 + 4x - 15 < 0$
Show clear algebraic working.

.....
(Total for Question 19 is 3 marks)

Worked Solution 29

Quadratic inequality

UNIT 2

3 marks

Factorise

$$4x^2 + 4x - 15 = (2x - 3)(2x + 5).$$

Use the sign of an upward quadratic

The expression is negative between the roots $-\frac{5}{2}$ and $\frac{3}{2}$.

Final answer

$$-2.5 < x < 1.5$$

Question 21

4MA1-2H Q15

ORIGINAL QUESTION

3 marks

- (b) Solve the inequality $3y^2 + 4y - 32 > 0$
Show your working clearly.

.....

(3)

Worked Solution 21

Quadratic inequality

MODULAR UNIT 2

3 marks

Factorise

$$3y^2 + 4y - 32 = (3y - 8)(y + 4).$$

Use the outside intervals

$$(3y - 8)(y + 4) > 0 \Rightarrow y < -4 \text{ or } y > \frac{8}{3}.$$

Final answer

$$y < -4 \text{ or } y > \frac{8}{3}$$

Question 04

4MA1-1H Q8 (a)

ORIGINAL QUESTION UNIT 2

2 marks

- 8 (a) Solve the inequality $8x - 4 \geq 3x - 10$

.....
(2)

Worked solution 04

Solve the linear inequality

MODULAR UNIT 2

2 marks

(a) – Solve the linear inequality

Collect terms

$$8x - 4 \geq 3x - 10 \Rightarrow 5x \geq -6.$$

Divide by positive 5

$$x \geq -\frac{6}{5}.$$

Final answer

$$x \geq -6/5$$

Original question 07

4MA1-1H Q8 M03

MODULAR UNIT 2

3 marks

(b) Solve the inequality $3y + 5 > 7y - 10$

Show clear algebraic working.

.....
(3)

Worked solution 07

Solve a linear inequality

MODULAR UNIT 2

3 marks

M03 – Solve a linear inequality**Collect terms**

$$3y + 5 > 7y - 10 \Rightarrow 15 > 4y.$$

Divide by a positive number

$$y < 15/4.$$

State the interval

$$y < 3.75.$$

Final answer

$$y < \frac{15}{4}$$

Original question 33

4MA1-2H Q20 M01

MODULAR UNIT 2

3 marks

- 20 Solve the inequality $10x^2 + 11x - 21 < 0$
Show clear algebraic working.
-

Worked solution 33

Solve a quadratic inequality

MODULAR UNIT 2

3 marks

M01 – Solve a quadratic inequality**Factorise**

$$10x^2 + 11x - 21 = (10x + 21)(x - 1).$$

Find critical values

$$x = -21/10 = -2.1 \text{ or } x = 1.$$

Choose the interval

The upward-opening quadratic is negative between the roots: $-2.1 < x < 1$.

Final answer

$$-2.1 < x < 1$$

Original question 28

4MA1-2H Q15 Q15(b)

ORIGINAL QUESTION**3 marks**

- (b) Solve the inequality $2y^2 - 7y - 30 \leq 0$
Show your working clearly.

(3)

Worked solution

Family: Solve quadratic inequalities • Variant: Quadratic inequality with inclusive roots

METHOD / WORKING

MODULAR UNIT 2 • JUNE 2022 DETAILED SOLUTIONS
Worked solution 28 MODULAR UNIT 2
Chapter: Solving Inequalities 3 marks
Q15(b) – Chapter: Solving Inequalities
Family: Solve quadratic inequalities • Variant: Quadratic inequality with inclusive roots
Method
$2y^2 - 7y - 30 = (2y + 5)(y - 6)$.
Why: Factorise the quadratic.
Method
The critical values are $y = -2.5$ and $y = 6$.
Why: Set each factor equal to zero.
Method
The quadratic opens upwards, so it is at most zero between the roots: $-2.5 \leq y \leq 6$.
Why: Select the interval where the product is non-positive and include both roots.
Final answer
solution set: $-2.5 \leq y \leq 6$

FINAL ANSWER

Original question 06

4MA1-1H Q6 Q6(a)

ORIGINAL QUESTION**2 marks**

6 (a) Solve the inequality $5x - 7 \leq 2$

(2)

Worked solution 06

Chapter: Solving Inequalities

MODULAR UNIT 2

2 marks

Q6(a) – Chapter: Solving Inequalities

Family: Solve linear inequalities • Variant: One-sided linear inequality

Method $5x - 7 \leq 2$ gives $5x \leq 9$.*Why: Add 7 to both sides.***Method** $x \leq 9/5 = 1.8$.*Why: Divide by the positive number 5.***Final answer****solution:** $x \leq 1.8$

Original question 34

4MA1-2H Q22

ORIGINAL QUESTION**5 marks**

22 Here is a rectangle.

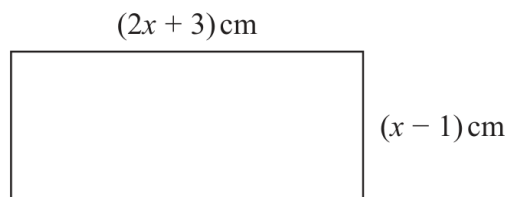


Diagram **NOT**
accurately drawn

Given that the area of the rectangle is less than 75 cm^2

find the range of possible values of x

Worked solution 34

Chapter: Solving Inequalities

MODULAR UNIT 2

5 marks

Q22 – Chapter: Solving Inequalities

Family: Solve quadratic inequalities • Variant: Contextual quadratic feasible range

Method $(2x+3)(x-1) < 75$ gives $2x^2 + x - 78 < 0$.*Why: Expand and move 75 to the left.***Method**The critical values solve $2x^2 + x - 78 = 0 = (x-6)(2x+13)$, so $x=6$ or $x=-6.5$.*Why: Factorise the boundary equation.***Method**Because the rectangle side $x-1$ must be positive, $x > 1$.*Why: Lengths are positive.***Method**The quadratic is negative between its roots, so $x < 6$.*Why: Use the sign of the upward-opening quadratic.***Method**Therefore $1 < x < 6$.*Why: Combine the restrictions.***Final answer**range: $1 < x < 6$

7 (a) Solve the inequality $2x + 7 > 4$

(2)

Worked solution

January 2020 | Question 7 | 2 marks

Family: Solve linear inequalities • Variant: One-sided linear inequality

Method / working

1. $2x+7>4$ gives $2x>-3$.

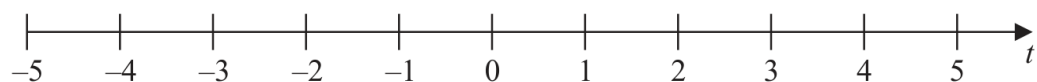
2. Divide by 2: $x>-3/2$.

Final answer: $x>-3/2$

(c) (i) Solve the inequality $7t - 8 < 2t + 7$

(2)

(ii) On the number line below, represent the solution set of the inequality solved in part (c)(i)



(1)

Worked solution

January 2021 | Question 5 | 2 marks

Family: Solve linear inequalities • Variant: One-sided linear inequality

Method / working

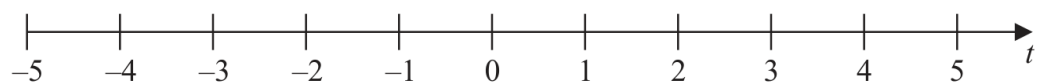
1. $7t - 8 < 2t + 7$ gives $5t < 15$.
2. Divide by 5 to obtain $t < 3$.

Final answer: $t < 3$

(c) (i) Solve the inequality $7t - 8 < 2t + 7$

(2)

(ii) On the number line below, represent the solution set of the inequality solved in part (c)(i)



(1)

Worked solution

January 2021 | Question 5 | 1 marks

Family: Translate inequality notation and number lines • Variant: Number-line endpoint convention

Method / working

1. Represent $t < 3$ with an open circle at 3 and shading to the left.

Final answer: Open circle at 3, arrow to the left

7 $-4 \leq 2y < 6$

y is an integer.

(a) Write down all the possible values of y .

(2)

Worked solution

June 2021 | Question 7 | 2 marks

Family: Translate inequality notation and number lines • Variant: Integers inside an inequality interval

Method / working

1. Solve $-4 \leq 2y < 6$ by dividing by 2: $-2 \leq y < 3$.
2. List the integer values: -2, -1, 0, 1, 2.

Final answer: -2, -1, 0, 1, 2

(b) Solve the inequality $7t - 3 \leq 2t + 31$

Show your working clearly.

(2)

Worked solution

June 2021 | Question 7 | 2 marks

Family: Solve linear inequalities • Variant: Linear inequality with unknown on both sides

Method / working

1. Collect terms: $7t - 3 \leq 2t + 31$ gives $5t \leq 34$.

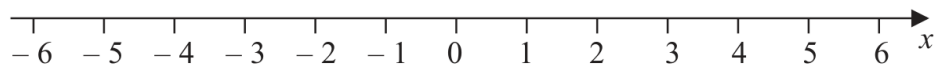
2. Divide by 5: $t \leq 34/5 = 6.8$.

Final answer: $t \leq 6.8$

9 (i) Solve the inequalities $-7 \leq 2x - 3 < 5$

(3)

(ii) On the number line, represent the solution set to part (i)



(2)

Worked solution

November 2021 | Question 9 | 3 marks

Family: Solve linear inequalities | Variant: Compound linear inequality

METHOD / WORKING

1. $-7 \leq 2x - 3 < 5$

2. Add 3 to each part:

3. $-4 \leq 2x < 8$

4. Divide by 2:

5. $-2 \leq x < 4$

6. On the number line, use a closed circle at -2 , an open circle at 4 , and shade between them.

7. Answer: $-2 \leq x < 4$.

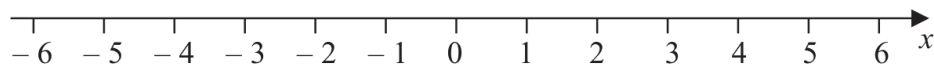
FINAL ANSWER

$-2 \leq x < 4.$

9 (i) Solve the inequalities $-7 \leq 2x - 3 < 5$

(3)

(ii) On the number line, represent the solution set to part (i)



(2)

Worked solution

November 2021 | Question 9 | 2 marks

Family: Translate inequality notation and number lines | Variant: Number-line endpoint convention

METHOD / WORKING

1. $-7 \leq 2x - 3 < 5$

2. Add 3 to each part:

3. $-4 \leq 2x < 8$

4. Divide by 2:

5. $-2 \leq x < 4$

6. On the number line, use a closed circle at -2 , an open circle at 4 , and shade between them.

7. Answer: $-2 \leq x < 4$.

FINAL ANSWER

$-2 \leq x < 4.$

1 (a) Simplify $g^6 \times g^4$

(1)

(b) Simplify $k^{10} \div k^3$

(1)

(c) Simplify $(3cd^4)^2$

(2)

(d) Solve the inequality $4x + 7 > 2$

(2)

(Total for Question 1 is 6 marks)

Answer reference | May-Nov 2020 | Question 1

Family: Solve linear inequalities • Variant: One-sided linear inequality

Q	working	ANSWER	MARK	NOTES
1 a		g^{10}	1	B1
b		k^7	1	B1
c		$9c^2d^8$	2	B2 B1 for 2 out of 3 terms correct in a product
d	$4x > 2 - 7$ oe			M1 accept as an equation or with wrong inequality sign.
		$x > -1.25$	2	A1 oe allow $(-1.25, (+)\infty)$ Note: award M1A0 for an answer on the answer line of -1.25 with no sign or the incorrect sign eg $x = -1.25$, $x < -1.25$
				Total 6 marks

Worked solution

May-Nov 2020 | Question 1 | 6 marks

Family: Solve linear inequalities • Variant: One-sided linear inequality

Official final mark-scheme pages are embedded immediately after this question.

11 (a) Simplify fully $(8e^{15})^{\frac{2}{3}}$

.....
(2)

(b) Express $\left(\frac{y}{2}\right)^{-4}$ in the form ay^n where a and n are integers.

.....
(2)

(c) Solve $\frac{4x-2}{3} - \frac{5-3x}{4} = 6$

Show clear algebraic working.

$x =$
(4)

(Total for Question 11 is 8 marks)

Answer reference | May-Nov 2020 | Question 11

Family: Translate inequality notation and number lines • Variant: Inequality symbol from a number line

11	a		$4e^{10}$	2	B2 (B1 for $4e^k$ or ke^{10})
	b	A correct first step eg $\frac{y^{-4}}{2^{-4}}$ or $\left(\frac{y^4}{16}\right)^{-1}$ or $\frac{y^{-4}}{0.0625}$ or $\left(\frac{2}{y}\right)^4$ or $\frac{16}{y^4}$ or $\left(\frac{1}{y/2}\right)^4$ or $\frac{1}{\left(\frac{y}{2}\right)^4}$			M1 or for $16y^p$ where $p \neq -4$
			$16y^{-4}$	2	A1
	c	eg $12 \times \frac{4x-2}{3} - 12 \times \frac{5-3x}{4} = 12 \times 6$ or eg $4(4x-2) - 3(5-3x) = 12 \times 6$ or eg $\frac{4(4x-2)}{12} - \frac{3(5-3x)}{12} (=6)$ or eg $\frac{4(4x-2)-3(5-3x)}{12} (=6)$ oe			M1 for clear intention to multiply all terms by 12 or a multiple of 12 or to express LHS as two fractions over 12 or a multiple of 12 or as a single fraction with a denominator of 12 or a multiple of 12 (if expanded numerator, allow one sign error)
		eg $16x-8-15+9x=6 \times 12$			M1 expanding brackets and multiplying both sides by denominator with no more than one sign error
		eg $16x+9x=72+8+15$			M1 for correct rearrangement of a correct equation with terms in x isolated
			3.8	4	A1 oe, award full marks for a correct answer if at least M1 scored
Total 8 marks					

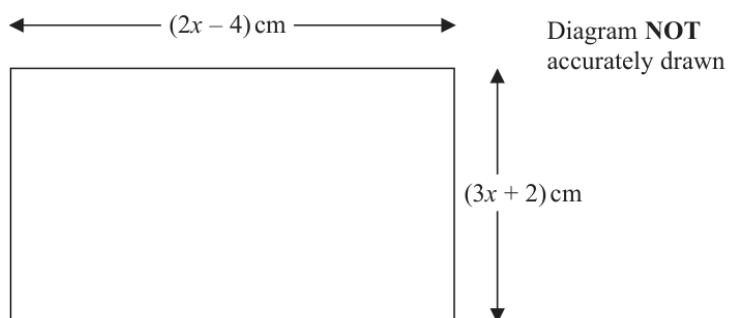
Worked solution

May-Nov 2020 | Question 11 | 8 marks

Family: Translate inequality notation and number lines • Variant: Inequality symbol from a number line

Official final mark-scheme pages are embedded immediately after this question.

17 The diagram shows a rectangle.



The area of the rectangle is $A \text{ cm}^2$

Given that $A < 3x + 27$

find the range of possible values for x .

(Total for Question 17 is 5 marks)

Answer reference | May-Nov 2020 | Question 17

Family: Translate inequality notation and number lines • Variant: Inequality symbol from a number line

17	$(3x+2)(2x-4) < 3x+27$ or eg $6x^2 - 8x - 8 < 3x + 27$ eg $6x^2 - 11x - 35 < 0$			M1 condone incorrect symbol
				M1 expanding and rearranging to get a correct 3 term quadratic, condone incorrect symbol
	$(2x-7)(3x+5) (= 0)$ or $\frac{11 \pm \sqrt{(-11)^2 - 4 \times 6 \times (-35)}}{2 \times 6}$			M1 first step to find the critical values dep on M1 for solving their 3 term quadratic using any correct method (allow one sign error and some simplification – allow as far as the equivalent of $\frac{11 \pm \sqrt{121 + 840}}{12}$) or if factorising, allow brackets which expanded give 2 out of 3 terms correct)
	$-\frac{5}{3}, \frac{7}{2}$			A1 oe the positive critical value only or both critical values (if both they must be correct)
		$2 < x < \frac{7}{2}$	5	A1 accept $2 \leq x < \frac{7}{2}$ may be seen as two separate inequalities $x > 2$ ($x \leq 2$) and $x < \frac{7}{2}$
				Total 5 marks

Worked solution

May-Nov 2020 | Question 17 | 5 marks

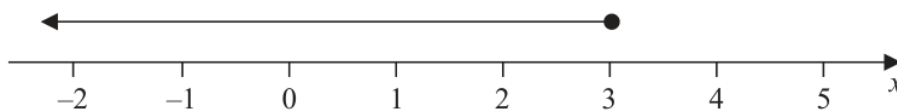
Family: Translate inequality notation and number lines • Variant: Inequality symbol from a number line

Official final mark-scheme pages are embedded immediately after this question.

10 (a) Factorise $y^2 - 2y - 48$

(2)

(b) Write down the inequality shown on the number line



(1)

(c) Solve the inequality $7w + 6 > 12w + 14$

(3)

(Total for Question 10 is 6 marks)

Answer reference | January 2023 | Question 10

Family: Translate inequality notation and number lines • Variant: Inequality symbol from a number line

10	(a)	$(y \pm 6)(y \pm 8)$ or $y(y+6)-8(y+6)$ or $y(y-8)+6(y-8)$		2	M1 or for $(y \pm a)(y \pm b)$ where $ab = -48$ or $a + b = -2$
			$(y+6)(y-8)$		A1 oe Allow any letter for y
	(b)		$x \leq 3$	1	B1 allow $3 \geq x$ Allow any letter for x
	(c)	$6-14 > 12w-7w$ oe or $7w-12w > 14-6$ oe		3	M1 Condone = rather than > or any other sign for this mark.
		$-8 > 5w$ or $-5w > 8$ or $-w > \frac{8}{5}$ or $w > -\frac{8}{5}$ or $w = -\frac{8}{5}$ oe			M1 Condone = rather than > or any other sign for this mark.
		<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	$w < -\frac{8}{5}$		A1 oe accept $-\frac{8}{5} > w$ Must have correct sign on answer line dep on M1 (sight of correct answer in working space and just $(w =) -\frac{8}{5}$ oe on answer line gains M2 only)
					Total 6 marks

Worked solution

January 2023 | Question 10 | 6 marks

Family: Translate inequality notation and number lines • Variant: Inequality symbol from a number line

Official final mark-scheme pages are embedded immediately after this question.

- 22 Solve the inequality $6x^2 + 37x \leq 35$
Show clear algebraic working.

.....
(Total for Question 22 is 3 marks)

Answer reference | January 2023 | Question 22

Family: Solve linear inequalities • Variant: One-sided linear inequality

Question	working	Answer	Mark	Notes
22	$(6x - 5)(x + 7)(=0) \text{ or }$ $\frac{-37 \pm \sqrt{37^2 - 4 \times 6 \times -35}}{2 \times 6}$ $6 \left[\left(x + \frac{37}{12} \right)^2 - \left(\frac{37}{12} \right)^2 \right] \dots \text{ oe}$		3	M1 A correct method to solve the quadratic equation $6x^2 + 37x - 35(=0)$ using any correct method (if factorising, allow brackets which expanded give 2 out of 3 terms correct) (if using formula allow one sign error in substitution and some simplification – allow as far as $\frac{-37 \pm \sqrt{1369 + 840}}{12}$) or completing the square as far as shown on left
	$\frac{5}{6}$ oe and -7			A1 dep on M1 correct critical values (allow 0.83...)
	<i>Working must be seen for both accuracy marks as asked for in question</i>	$-7 \leq x \leq \frac{5}{6}$		A1 dep on M1 oe eg $-7 \leq x \leq 0.83\dots$, $\left[-7, \frac{5}{6} \right]$ Accept $x \leq \frac{5}{6}, x \geq -7$
Total 3 marks				

Worked solution

January 2023 | Question 22 | 3 marks

Family: Solve linear inequalities • Variant: One-sided linear inequality

Official final mark-scheme pages are embedded immediately after this question.

- 25 Solve the inequality $2x^2 + x - 28 > 0$
Show clear algebraic working.

(Total for Question 25 is 3 marks)

Worked solution

November 2025 | Question 25 | 3 marks

Family: Translate inequality notation and number lines | Variant: Inequality symbol from a number line

METHOD / WORKING

Worked solution

November 2025 | Question 25 | 3 marks

Family: Translate inequality notation and number lines | Variant: Inequality symbol from a number line

Method

$$2x^2 + x - 28 > 0$$

Factorise:

$$2x^2 + x - 28 = (2x - 7)(x + 4)$$

The critical values are

$$x = \frac{7}{2} \text{ and } x = -4$$

The quadratic is greater than 0 outside the roots.

Answer: $x < -4$ or $x > \frac{7}{2}$

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER